particles in the atmosphere. Nevertheless, these drawbacks are not insurmountable. Over the course of many decades, experimenters have learned quite a lot from studying the more plentiful, lower-energy cosmic ray data, and, to deal with the paucity of high-energy collisions, experimenters have built huge arrays of detectors to catch as many particles as possible.

7. The expert reader will realize that conservation of energy in a theory with dynamic spacetime is a subtle issue. Certainly, the stress tensor of all sources for the Einstein equations is covariantly conserved. But this does not necessarily translate into a global conservation law for energy. And with good reason. The stress tensor does not take account of gravitational energy—a notoriously difficult notion in general relativity. Over short enough distance and time scales—such as occur in accelerator experiments—local energy conservation is valid, but statements about global conservation have to be treated with greater care.

8. This is true of the simplest inflationary models. Researchers have found that more complicated realizations of inflation can suppress the production of gravitational waves.

9. A viable dark matter candidate must be a stable, or very long-lived, particle—one that does not disintegrate into other particles. This is expected to be true of the lightest of the supersymmetric partner particles, and hence the more precise statement is that the lightest of the zino, higgsino, or photino is a suitable dark matter candidate.

10. Not too long ago, a joint Italian-Chinese research group known as the Dark Matter Experiment (DAMA), working out of the Gran Sasso Laboratory in Italy, made the exciting announcement that they had achieved the first direct detection of dark matter. So far, however, no other group has been able to verify the claim. In fact, another experiment, Cryogenic Dark Matter Search (CDMS), based at Stanford and involving researchers from the United States and Russia, has amassed data that many believe rule out the DAMA results to a high degree of confidence. In addition to these dark matter searches, many others are under way. To read about some of these, take a look at *http://hepwww.rl.ac.uk/ukdmc/dark_matter/other_searches.htm*.